

Experiment 3: Loan Sanction Amount Prediction using Linear Regression Models

1 Aim

- To build regression models to predict the Loan Sanction amount based on property and financial attributes.
- To compare multiple regression models and their performance.

2 Dataset

The loan prediction dataset is split into train and test split already. It consists of 30,000 records with 24 feature columns:

- **Identifiers:** Customer ID, Name, Property ID
- **Demographics:** Gender, Age, Dependents
- **Financial attributes:** Income (USD), Income Stability, Profession, Type of Employment, Current Loan Expenses (USD), Credit Score, No. of Defaults, Has Active Credit Card
- **Loan attributes:** Loan Amount Request (USD), Expense Type 1, Expense Type 2
- **Property attributes:** Property Age, Property Type, Property Location, Property Price, Co-Applicant, Location
- **Target variable:** Loan Sanctions

There are many columns with missing values including Income, type of employment, property age, credit score etc. The target column has 340 missing values and is right skewed.

3 Preprocessing Steps

- Customer ID and Name were dropped.
- Missing target value rows were removed.
- Missing values were imputed using median.

4 Implementation

- `LinearRegression` was fit on the scaled training data.
- Ridge, Lasso and ElasticNet each fit and tuned.
- `GridSearchCV` with 5 fold CV was used to search.

Each model's RMSE was recorded for overfitting analysis.

5 Visualization

Figure 1 presents the target distribution, feature – target scatter plot, predicted vs actual values and a coefficient comparison bar plot across all models.

All models perform very similarly, resulting that regularization has less effect on the data once the features are scaled. Overfitting is not an issue here.

The models show underfitting – the linear model is too simple to capture the relation between the features.

6 Bias–Variance

High Bias – Low Variance: All model converge to a similar error, therefore they are stable but cannot show full feature relation due to linear model.

7 Conclusion

- Feature scaling was required for Elastic Net and Ridge regression to converge.
- Loan Amount Request is one of the strongest feature to predict the target.
- All models have similar RMSE.
- Regularization does not have an impact.

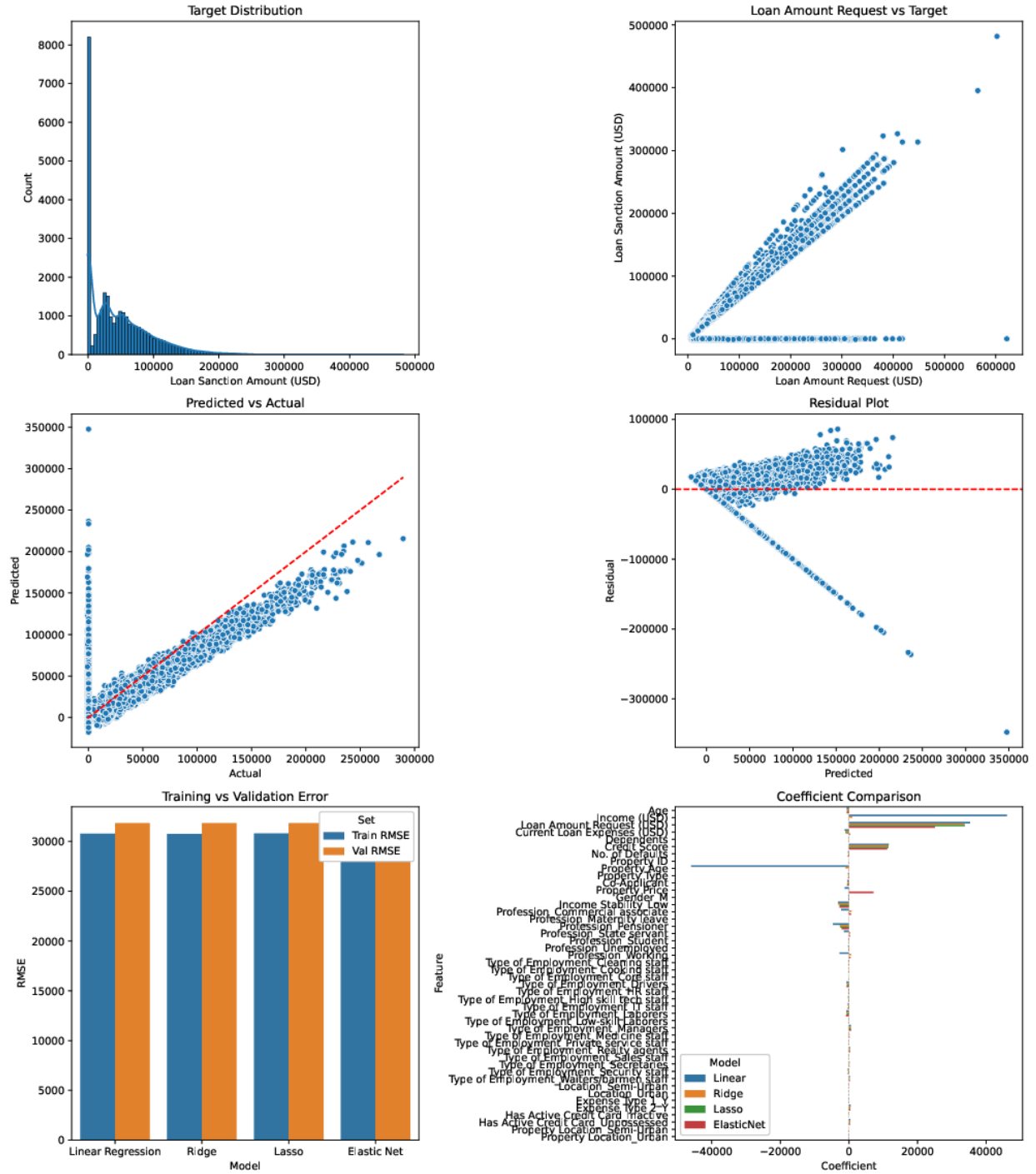


Figure 1: Combined visualization panel: (top-left) target distribution, (top-right) feature vs. target scatter, (middle-left) predicted vs. actual, (middle-right) residuals, (bottom-left) training vs. validation error, (bottom-right) coefficient comparison across models.